Predicting User Intention as Next Direction in Redirected Walking

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Abstract. How to freely explore virtual space in a limited physical space is a technical challenge in virtual reality technology. In this paper, a new Redirected Walking (RDW) technique based on user intention prediction is proposed. By collecting the walking data of real users, the short-term future walking intention is inferred by utilizing the user's motion characteristics. Then physical trajectory planning is performed based on the user's intention to guide the user's walking. Simulation experiments show that the method can effectively reduce the number of user resets due to collisions with physical obstacles. Based on user experiments with 20 real users, it is shown that the method significantly reduces the number of collision resets while effectively avoiding the strong discomfort of the user in the Virtual Reality (VR) task.

Keywords: Virtual reality · Redirected walking · Interaction techniques · Intention prediction.

1 Introduction

With the breakthrough of hardware and software technology, Virtual Reality (VR) technology has been developed rapidly, which gradually changes the way of human-computer interaction and makes VR technology widely used in the fields of socialization, education, and medical treatment. In all kinds of VR tasks, movement is the most basic and common interaction method. Since the virtual space is generally much larger than the physical space, it is difficult to map the user's movement in the physical space to the virtual space in a 1:1 ratio. In order to enable users to walk freely in virtual space, many mobility techniques for VR have emerged, such as invisible teleportation, walking-in-place [9, 20, 28], omnidirectional treadmill [18], and Redirected Walking (RDW) [23, 25]. Among them, RDW is a virtual space roaming technique based on the natural walking of human beings, which mainly utilizes the user's uncertainty of perception to

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perform subtle manipulation of the virtual camera to change the mapping of the user's movement between the virtual space and the physical space without the user being able to notice it, with the aim of enabling the user to roam the virtual space as infinitely as possible in the limited physical space.

When applying the RDW algorithm to guide the user's walking, the user's physical trajectory becomes uncertain by the effect of the perceptual gain, which leads to the user's collision with physical obstacles becoming uncontrollable. Some studies analyze the physical place by using Artificial Potential Filed (APF) [27], or by precomputing the reachable poses (position and orientation) in the physical space, and then using the idea of trajectory planning to guide the user to avoid collisions [33]. In order to realize trajectory planning, a virtual target point needs to be introduced to align with the actual physical target point for subsequent trajectory planning steps. The use of intention prediction to calculate the user's future virtual position and thus achieve reliable trajectory planning is a more effective means.

The prediction class RDW algorithm [13] dynamically adjusts the perceptual gain by predicting the user's future virtual pose or walking trajectory to avoid collisions with obstacles in the physical space. The user's future virtual pose is determined by the user's intention, which is affected by the virtual environment in which the user is located. Due to the variability of the virtual environment, if the environment data is directly utilized to assist in determining the user's intention, the model needs to be retrained every time the virtual scene is changed. Some studies have attempted to improve the prediction accuracy by directly utilizing the user's intention, such as judging the user's intention through the user's eye movement data [24, 35], and combining the historical trajectory data to predict the user's future position [5]. However, such methods require the use of specific hardware devices, which increases the cost and the difficulty of data collection in practical applications. In this paper, we design a redirected walking method based on user's intention prediction, which predicts the short-term future walking intention by utilizing the user's motion characteristics, and then calculates the user's virtual position, matches this virtual position with the physical position, and realizes the planning of the user's future trajectory to be used for redirected walking. The method uses an ordinary VR headset to collect the user's walking data, which significantly reduces the requirements for hardware devices.

In summary, the contribution points of this paper can be summarized as:

1 A user intention prediction method that can be used in ordinary VR headsets is designed, which improves the applicability of prediction-based RDW algorithms.
2 We proposed an intention prediction-based RDW algorithm, which is able to speculate the user's future virtual position according to the user's intention, and then align with the standard physical position for trajectory planning to guide the user's walking, which solves the problem of the uncertainty of the user's collision with physical obstacles.

3 The algorithm effectively reduces the number of resets generated by the user's collision with physical obstacles, and at the same time effectively avoids the user's strong discomfort in the VR task.

2 Related work

The concept of redirected walking algorithms [23] was first introduced in 2001 by Razzaque et al. Redirection-aware gain is the basic way to realize redirected walking, where inconsistencies between the user's virtual and physical motion states are achieved through subtle manipulations. Existing research works mainly classify the gains into three categories, including translation gain, rotation gain, and curvature gain [21], and these redirection gains realize the redirection control of the user by changing the user's moving speed, body rotation, and moving direction. By using different redirection gains in reasonable combinations, many redirection walking methods have been proposed to realize the user's realistic walking in a wide range of virtual spaces. However, due to the limitation of the redirection-aware gain threshold, the redirection walking technique is limited in the motion deviation that can be imposed between the user's virtual and physical motions, and the user will still inevitably collide with boundaries and obstacles during roaming, and at the same time, due to the use of gains, the collision has a great deal of uncertainty, and at present, the following types of algorithms are able to ameliorate this problem to some extent.

2.1 RDW algorithm based on physical environment analysis

Since heuristic strategies in the past were fixed, they usually could not solve the problem of complex distribution of obstacles in physical space well. Thomas et al. generated a potential field for the whole physical environment in virtual space based on the idea of APF and assumed that places with obstacles have repulsive force on the user, pushing the user to the open space to avoid collision [27]. Later there are some approaches based on reinforcement learning that use neural networks to generate suitable physical goals for the user to guide the user to walk. Chen et al. proposed two greedy strategies, steer-to-farthest and trapezoidal roadmap, which enable the RDW algorithm to adapt to irregular physical environments [4]. Some subsequent algorithms guide the user to a specific physical location and enhance the user-environment interaction through passive tactile feedback or visual cues [38, 14, 17]. Later Xu et al. used predefined arcs to plan physical paths by analyzing the physical environment [31], but this algorithm lacks some flexibility and is prone to collide with obstacles. Based on this they proposed a new method to navigate the user to a specific physical location using curvature gain[33] , but this method only generates a path and does not completely avoid obstacles, and changing the direction of the curvature gain several times increases the user's sense of vertigo, and at the same time, the above algorithm does not take into account the influence of the virtual environment when calculating a specific physical target point. By adding the consideration

of virtual targets and making each reset as far away from the virtual target as possible, they succeeded in reducing the number of resets generated by the user's collision with physical obstacles [32].

2.2 Prediction-based RDW algorithm

The effectiveness of redirection algorithms based on path prediction can be significantly improved when the accuracy of the user's future virtual path prediction is high. Such algorithms applying predicted path outcomes can be categorized into two groups, short-distance path prediction and long-distance path prediction [22]. Nitzsche et al. first used the user's head-tracking data to obtain the user's expected walking direction, which can provide reliable path prediction within the next few seconds [19]. However, since the user's exploration in virtual space is irrational, in order to solve this problem, previous approaches restricted the layout of the virtual environment to narrow passages to ensure the accuracy of the predicted trajectories obtained. Hirt et al. proposed a new RDW technique for short-term prediction to address this problem [10], which introduces a teardrop shaped trajectory prediction algorithm using Bernoulli bi-nuclei representation to enable it to be application to empty virtual environments. In contrast long-term prediction requires the use of the user's virtual goal information to aid in prediction, i.e., it needs to utilize information from the virtual environment. Zank proposed a skeleton graph search algorithm for redirection planning and potential goal path point prediction [36], which is able to apply previous prediction-type algorithms to large-scale virtual environments. A 2016 study by Gandrud and Interrante found that a user's future directional intention can be obtained from head movements and eye gaze data during motion, providing a theoretical basis for subsequent user trajectory prediction using eye movement data [7]. Subsequent works have introduced neural networks and recurrent networks to improve the prediction of user trajectories by using eye-movement information as auxiliary information [2, 24]. Jeon proposed a new algorithm that uses an LSTM model with a combination of the user's spatial information and eye-movement tracking data, to predict the user's future position in virtual space without any assumptions about the preconditions, and to apply these predictions are applied to existing RDW methods [13], a mechanism that significantly reduces the number of resets and increases the distance between resets, improving the effectiveness of existing RDW algorithms.

2.3 Intention Prediction Method Based on Pedestrian Posture

Currently, research related to modeling pedestrian intentions is mainly in the field of autonomous driving. Recent studies have shown that the pedestrian's movements (waving, gazing, etc.) before crossing the road are related to the person's walking intention (whether to cross the road or not). Pedestrian's pose is highly correlated with his/her gait and associated with other data such as pedestrian's head orientation and body orientation. Earlier approaches were mostly based on Convolutional Neural Networks (CNN) [34] and LongShort

Term Memory (LSTM) [8] using human skeleton features for action recognition and trajectory prediction, e.g., FangZ et al. proposed monocular vision-based human pose estimation method [6] to predict the intention of pedestrians and cyclists.Zhang S et al. found that using geometric relationship features based on the distance between joints and selected lines outperforms other features [37] and provided a simple general spatial modeling method perpendicular to RNN model enhancement, which further improves the model performance. Recent studies have started to introduce graph structures in human pose estimation tasks, for example, Cadena P et al. developed a 2D pedestrian graph structure and a pedestrian graph network to predict whether a pedestrian is going to cross the road or not [3]. Huynh M et al. proposed the GPRAR model [12], which is based on a graph convolutional network for human pose reconstruction and action recognition, and the model consists of a feature aggregator, the feature aggregation The model includes a feature aggregator, which aggregates learned features such as human pose, action, and position in a channelized manner, and uses an encoder-decoder based convolutional neural network to predict future positions.

Although these methods can predict the intention of pedestrians crossing the road more accurately, the data used for training and prediction are the human skeleton feature data or the image data of a person while walking, which makes the network more demanding on computer arithmetic and more redundant information, and it is not suitable for application on VR headsets. In addition, in the RDW scenario, the user's walking intention is not only whether to cross the road or not, but also contains more complex situations such as turning and stopping. Therefore, this paper attempts to predict more complex user intentions by collecting and analyzing motion data such as user's head direction and walking speed while walking based on the existing algorithms.

3 Methods

3.1 Idea validation

Data collection This paper uses Unity 2021.3.25f to build a 10m*15m simulation scene. The simulation scene is a simple painting exhibition with six paintings and two resting benches. In order to collect enough user motion data, seven target positions are set according to the arrangement of the paintings and the positions of the resting benches, and the user needs to follow the prompts to walk to these target positions in a certain order to complete seven kinds of actions: walk straight, turn left/right, pause, turn head to right/left and turn around.

The specific layout of the painting exhibition and the user's walking order are shown in Fig. 1, in which the blue rounded squares represent "paintings", the gray rounded squares represent "resting benches", i.e., obstacles, the red circle represents the target location, and the numbers in the circle are the order in which they appear. The red circles represent the target locations, and the numbers in the circles indicate the order in which they appear. Users will be prompted to "walk towards the red circle" before the start of each mission. When

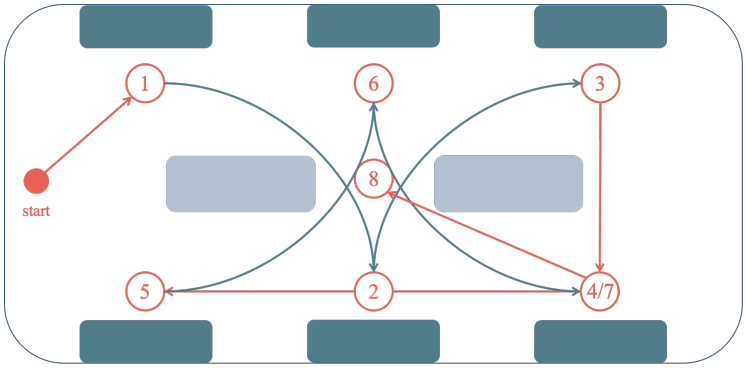


Fig. 1: The layout of the painting exhibition and the user's walking sequence. In the diagram, blue squares represent the "paintings," yellow squares represent the "resting benches," and red circles indicate the target locations, with numbers showing the order in which they appear. At the start of each task, users are prompted to "walk toward the red circle." Once they reach a target circle, the circle turns green, and users are then prompted to walk to the next red circle. The red paths on the diagram indicate straight walking behavior, while the blue paths represent turning movements. Upon reaching the final target circle (circle 8), users are asked to look at a randomly highlighted painting, which triggers head-turning behavior.

the user reaches the position corresponding to a red circle, the circle turns green and the user is prompted to walk to the next red circle. The red path in the image corresponds to the user's straight forward behavior, the blue path corresponds to the user's turning behavior, and walking to the red circle but not reaching the center of the circle corresponds to the user's stopping behavior. When the user walks to the final target circle (i.e., circle 8 in the figure), he/she needs to look at a randomly highlighted painting, which corresponds to the user's head-turning behavior. If the highlighted painting is located behind the user, it corresponds to the user's turning behavior. The process data of various types of user behaviors are recorded and labeled.

Real motion characterization data were collected using the Oculus Quest 2 from 10 users, 5 males and 5 females, with an age distribution between 22-25, 9 with myopia, and 5 who had used a VR headset. Each user took an average of 2 minutes to complete a full task, and user motion data were collected at a frequency of 20 Hz.

The data collected had a total of 6 observations, which were the coordinates of the user's position in the virtual space, the rotation angle of the user's head, and the need to standardize the user's positional coordinates in order to reduce the influence of a specific scenario on the data. The user's relative position in the whole scene is obtained as the final positional feature data by dividing the

user's positional coordinates collected at each time by the length and width of the virtual scene, respectively.

The user's walking speed, acceleration, rotational angular velocity, the change in position coordinates between every two acquisition time points, and the change in rotation angle were additionally computed using six observations and an acquisition time interval (0.05s). These nine computed values, together with the six observations, form a 15-dim vector that serves as the user motion characterization data.

Problem modeling In this paper, a network model containing three fully connected layers is constructed for extracting the relationship between motion data and user intention. The first two fully-connected layers of the network use the relu activation function with a cross-entropy loss function and Adam optimizer with a learning rate of 0.001, a total of 20 rounds of training, and a batch size of 128. Considering that the walking speed of a real user while wearing the headset is about 1m/s, the model predicts the results for user intentions after 3s. Since turning and stopping behaviors do not cause collisions with physical obstacles, the model predicts only the remaining five intentions.

Data processing The collected behavioral data of 10 users are divided into training set, validation set and test set in the ratio of 3:1:1. The metrics used for evaluation are precision, recall, and F1-score, where precision is the proportion of samples predicted by the model to be positive cases that are actually positive cases, which measures the accuracy of the model in the samples predicted to be positive cases. Recall is the proportion of samples correctly predicted as positive by the model to the actual number of positive samples, which measures the model's detection rate of positive samples.F1-score is the harmonic mean of precision and recall, which combines the model's accuracy and detection rate.F1-score is a commonly used composite performance indicator, especially for unbalanced category distributions.

Table 1: Prediction effect

State	Precision	Recall	F1-Score
Straight forward	0.9349	0.9497	0.9422
Turn head to left	0.8069	0.7857	0.7962
Turn head to right	0.6768	0.6537	0.6650
Turn left	0.8805	0.8995	0.8899
Turn right	0.9346	0.9340	0.9343

Data analysis Table 1 shows the model's prediction effect on five kinds of user walking intentions after 3s. From the table, it can be seen that the prediction

effect of walking straight, turning left and turning right is better, and the accuracy is basically above 0.85. It is worth noting that the accuracy of turning left and turning right is not high. By analyzing the collected data and observing the user's behavior during collection, it can be found that users have a certain preference for turning their heads when looking for highlighted paintings, in which there are more users who turn their heads to the left first to find paintings, so there is a certain chance that the user's behavior of turning their heads in both directions occurs at the same time during the collection of the data, which leads to the confusion of the two behaviors.

3.2 Applications of intentional prediction

Calculation of virtual postures After getting the user's future intention, it is necessary to use it for the calculation of virtual bit position. Assuming that the user's current position is (x_s, y_s, z_s), the user's direction is θ_s, the user's future position coordinates are (x_e, y_e, z_e), and the future direction is θ_e, then the user's future bit position can be expressed by Equation 1.

$$\begin{gathered} x_e = x_s + d * \cos(\theta_s) \\ y_e = y_s \\ z_e = z_s + d * \sin(\theta_s) \\ \theta_e = \theta_s + (d * c) \end{gathered} \tag{1}$$

which d represents the distance traveled by the user. c denotes the curvature of the circular trajectory that the user walks, and these two are assigned values according to different situations. When the user's future intention is to walk straight, there is:

$$\begin{gathered} d = v * \Delta t \\ \theta_e = \theta_s \\ c = 0 \end{gathered} \tag{2}$$

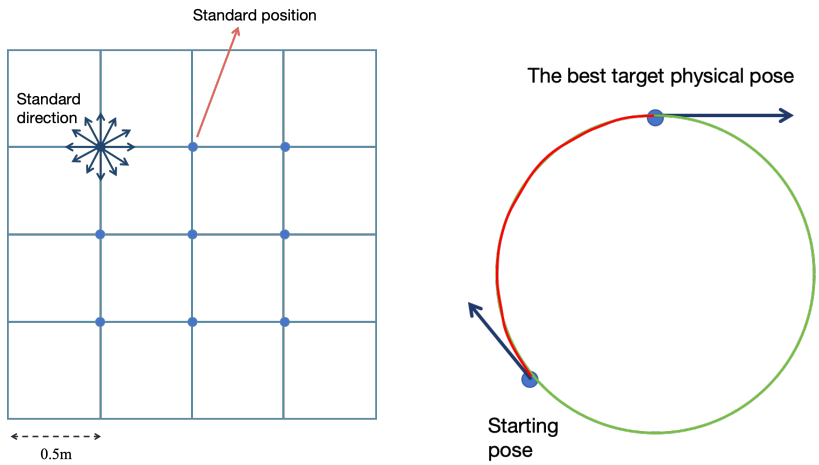
which v represents the speed at which the user is walking. When the user's future intention is to turn left or right, there:

$$\begin{gathered} d = v * \Delta t \\ \theta_e = d * c \\ c = \frac{1}{7.5} \end{gathered} \tag{3}$$

This c comes from the commonly used curvature gain threshold of 7.5m, which is represented by the fact that a user walking along a straight line in virtual space can at most walk along a circular arc with a radius of 7.5m in physical space without being aware of it.

Aligning virtual pose with physical pose After obtaining the virtual pose, it needs to be mapped into the physical space. Firstly, the distance between the virtual pose and the user's current position is calculated, and after obtaining

the distance, it is converted to a feasible distance in the physical space using the translation gain. The commonly used translation gain interval ranges from 0.86 to 1.26, which means that when the virtual distance is 1m, the user's walking distance in the physical space can be within the range of 0.86m to 1.26m.



(a) standard pose (b) path planning

Fig. 2: Schematic diagram of standard pose and path planning. As shown in (a) we sampled the physical space at 0.5-meter intervals to establish standard positions. Each of these positions was further divided into 12 directions at 30-degree intervals. In (b), the blue dot represents the pose point, the green circle shows the path circle with a radius of at least 7.5 meters, and the red line indicates the planned path.

Referring to Xu et al.'s method of discretely dividing the physical space into position points [33], the physical space is sampled at equal intervals every 0.5m to get the standard positions, and at the same time, each position is decomposed into 12 standard directions at equal intervals of 30 degrees, and finally, the uniformly distributed reachable position points in the physical space are obtained, as shown in Fig. 2a. After that, according to the physical distance obtained from the conversion and the target direction, we find the physical position point with suitable distance and small direction deviation as the target point.

When converting, the distance from the user's current position to the virtual target position is firstly calculated, and two values are obtained after converting using the translation gain, which are the maximum and minimum distances that the user can walk in the physical space, respectively. After that, the nearest physical location corresponding to the user's current location is found, the Euclidean distance between the current physical location and all standard physical locations is calculated, and the standard location with the smallest difference

between the distance value and the maximum distance or minimum distance is selected. If more than one standard location point meets the requirements, the one with the farthest distance from the physical location corresponding to the user's current location is selected. After the target location is selected, the target direction needs to be determined. Similar to the previous section, the direction with the smallest difference from the standard direction is selected as the target direction. A complete physical position point for trajectory planning can be obtained by the above steps.

Finally, an arc with a radius of 7.5m is used to connect the user's start point to the goal point [32] as the user's planning path.

As shown in Fig. 2b, the blue points are the position points, the green circle is the trajectory circle with a radius of not less than 7.5m, and the red color is the selected planning path. According to the radius of the trajectory circle, the trajectory curvature is calculated and converted into curvature gain, which is used to guide the user in the walking process.

4 Simulation experiment

4.1 Additional experimental details

Similar to many previous approaches [27, 1, 15, 30], in this paper, the speed of the avatar in the virtual space is set to 1 meter per second and the rotation rate is set to $\pi/2$ radians per second simulate the motion of a real user in the virtual space. Each path point used to guide the virtualized avatar to walk has a random angle in the virtual space ranging from $-\pi$ to π, and the distance between each path point is in the range of 2m-8m.

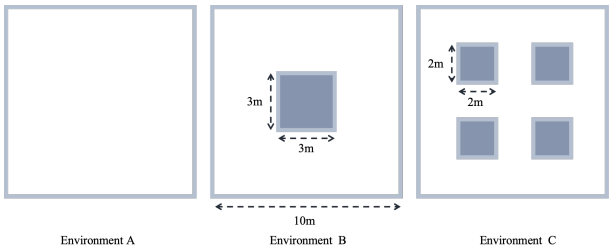
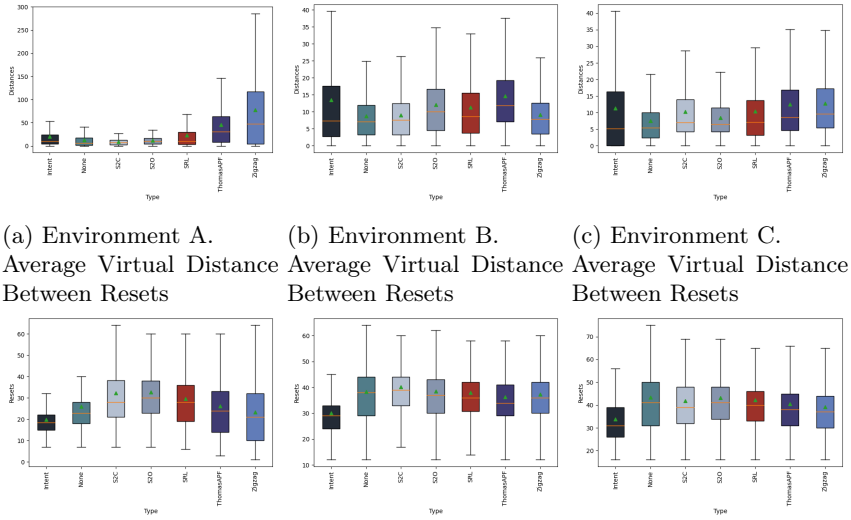


Fig. 3: Schematic diagram of path planning

The physical environments in the simulation experiments use three representative physical environments in the RDW algorithm, which are called environments A, B, and C. As shown in Fig. 3, the three environments are all squares with a side length of 10 m, in which there is no obstacle in environment A, there is an obstacle with a square with a side length of 3 m in the center of environment B, and there are four obstacles in environment C with a square with a



(a) Environment A. Average Virtual Distance Between Resets

(b) Environment B. Average Virtual Distance Between Resets

(c) Environment C. Average Virtual Distance Between Resets

(d) Environment A. Number of Resets

(e) Environment B. Number of Resets

(f) Environment C. Number of Resets

Fig. 4: The box plots of the results of experiments with random paths

side length of 2 m uniformly distributed in the the space. At the same time, a reset buffer with a width of 0.5m was set at all obstacles and the environment boundary.

The virtual environment was set as an open area without any objects. 100 random paths were generated in the whole area, each with 100 path points. Various comparison algorithms use the same 100 paths to exclude confounding factors in the experiment. The avatar needs to walk along each path until it touches all the path points.

The main evaluation metrics are the number of resets and the walking distance between two resets. The fewer the number of resets, the better the redirection controller, because fewer resets means fewer interruptions to the user's task, which is more conducive to safeguarding the user's immersion. The longer the distance between two resets, the better, the longer this distance represents the longer the distance the user can walk completely, also more conducive to safeguarding the user's sense of immersion.

In this paper, OpenRDW [16] is used to conduct simulation experiments, and the comparison algorithms include S2C [11], S2O [11], SRL [26], ThomasAPF [27], Zigzag [23], and a control group without algorithm interference. In case the user collides with a boundary or an obstacle, a reset is performed using the 2:1-turn method [29].

4.2 Experimental results and discussion

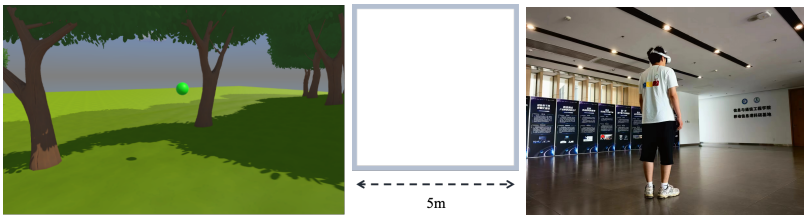
The distribution of the number of resets and the walking distance between every two resets for all the methods in different environment configurations are shown in Fig. 4. It can be observed that the methods in this paper show a relatively significant improvement in effectiveness compared to the existing algorithms in all three simulation experiment scenarios.

In simulation environment A, Intent presents a better effect, and its reset times are significantly reduced compared to None, S2C, S2O and SRL, and compared to ThomasAPF and Zigzag, although the effect is not much different, it is still less than these two algorithms, so Intent shows a clear advantage in the number of resets. Observing the distance between resets, it can be found that the Intent method proposed in this paper exceeds None, S2C and S2O, and is slightly worse than SRL, but there is no significant difference between Intent and SRL algorithms after analysis of variance ($P > 0.05$), and it can be assumed that the effects of both Intent and SRL are comparable.ThomasAPF is more effective in terms of the number of reset times than Intent based on the number of reset times, because it will push the user to the empty area constantly, compared with Intent based on the number of reset times. area, ThomasAPF is more adaptive than Intent in planning according to a specific location, and thus achieves better results.

For simulation environment B, from the analysis of the index of the number of resets, the only method with an average number of resets smaller than that of Intent is ThomasAPF, with an average number of resets of 27.74, which is found to have a low degree of significance after ANOVA analysis ($P > 0.05$), whereas for the other algorithms, Intent shows a clear advantage. From the metric of distance between two resets, for the rest of the algorithms except ThomasAPF, the distance between two resets is less than the Intent method. From the P-value obtained from the ANOVA, the difference between ThomasAPF and Intent is not highly significant ($P = 0.06$), and it can be assumed that Intent and ThomasAPF are equally effective. Therefore, in the simulation experiments in environment B, the Intent method is significantly better than the classical responsive algorithm and has comparable effects with the APF-based algorithm.

For simulation environment C, the overall distribution of the data shows that the number of resets goes up and the distance between resets goes down compared to the other two simulation environments. The reason for this is that the number of obstacles in the scene increases, leading to easier triggering of resets. In terms of the number of resets, the Intent method shows significant improvement compared to all types of existing methods. Analyzing the distance between resets, the Intent method is slightly less effective compared to the ThomasAPF algorithm, but the difference is not high ($P > 0.05$), and it can be considered as having a similar distance between resets. Compared with None method and classical responsive algorithm, there is a more significant improvement ($P < 0.05$).

5 User study



(a) Virtual environment (b) Physical environment layout (c) Test site photo

Fig. 5: The user study.

Experimental setup In order to further validate the effectiveness of the method proposed in this paper, 20 real users were summoned, including 12 males, 8 females, and 10 people who had previous experience of using VR. The experiment used OpenRDW to build the scene framework and Oculus Quest 2 as the user headset. The experiment created a forest virtual environment. During the experiment, users needed to keep walking towards the green ball to complete the task. A 5m*5m square obstacle-free space was used for the physical space. The specific virtual environment schematic and physical space layout are shown in Fig. 5, where the leftmost side is the virtual scene seen by the user, the middle is the layout of the physical space where the user is located, and the rightmost side is the scene record during the experiment.

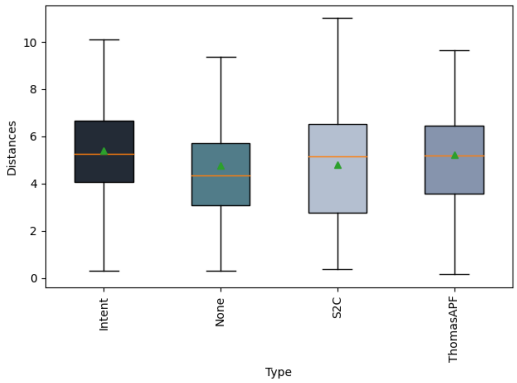
In the user's experiment, a total of 10 target points (green balls) were set up, the random distance between neighboring target points was between 2m-8m, and the target was generated with a random direction angle between to. It took the user approximately 5 minutes to complete each set of experiments. The algorithms used for control are None, S2C, and ThomasAPF, so in addition to the experimental group Intent, each user needs to complete three sets of control experiments. In order to verify whether the method in this paper causes motion sickness in users, users are asked to fill in the Simulator Sickness Questionnaire (SSQ) in the initial stage of the experiment to record their initial state. To ensure a fair analysis of the level of discomfort caused by each algorithm, users were asked to take a 2-minute break after each group of experiments and were asked to fill out the Simulator Sickness Questionnaire (SSQ) after each control and experimental group, and the effect of the group of experiments on the users was assessed by calculating the score of the questionnaire.

To assess the effect of the algorithms, the experiments recorded the number of resets and the distance between each two resets during the completion of the task and analyzed whether the users would experience discomfort during the completion of the task based on the SSQ scores.

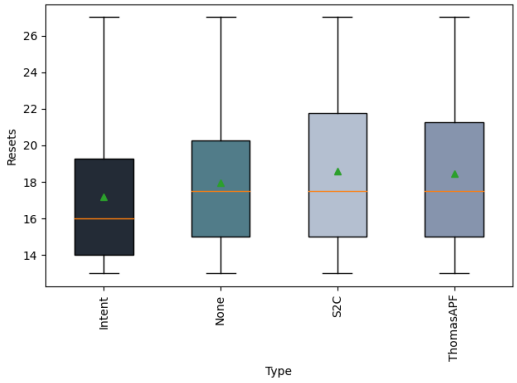
Results and discussion As shown in Fig. 6, in user experiments, the Intent method proposed in this paper presents better results than other compared

Table 2: SSQ-score. The first two columns represent the scores obtained from the SSQ questionnaires filled out by the users after completing the test group. The last two columns represent the name of the experimental group and the corresponding SSQ score, respectively.

Pre	Score	Type	Score
Test	8.27	None	7.28
		S2C	12.99
		ThomasAPF	29.14
		Intent(ours)	22.44



(a) Average Virtual Distance Between Resets



(b) Number of Resets

Fig. 6: The box plots of the results of experiments with 10 random waypoints.

methods. Between two resets, the Intent method allows the user to walk a longer distance and also reduces the number of resets generated by the user's collision with the boundary. After analyzing the SSQ questionnaire scores and user interviews, it can be learned that ThomasAPF causes the strongest motion sickness reaction to the user among the methods tested, due to the fact that the algorithm continuously changes the curvature gain value during the user's action, and even flips the direction of the curvature gain several times, resulting in a strong sense of discomfort for the user. In contrast, the Intent method aligns the virtual position with the physical position by predicting the user's intention, which meets the user's expectation when guiding the user to walk, and applies the curvature gain value that is the smallest value that satisfies the current situation, so the user does not have an overly strong feeling of discomfort. As can be seen from Table 2, the Intent method has lower SSQ scores compared to ThomasAPF, which proves that the Intent method does not have a drastic effect on the user in a short period of time.

6 Conclusion and Future Works

In this paper, we design a RDW algorithm based on the prediction of user's future intention, which analyzes the current user's motion data to predict the user's walking intention after 3s, converts the user's intention into the user's possible future virtual pose points after obtaining the user's intention, and then conducts the trajectory planning by aligning it with the standard physical pose points, and finally guides the user to walk. Both simulation experiments and real user experiments show that this method can effectively reduce the number of resets generated by the user's collision with boundaries or obstacles, while not bringing strong discomfort to the user.

In this paper, a shallow neural network is used for user intention prediction, and its performance is constrained by the richness of user data. Improving the intention prediction accuracy by exploring other more complex models may yield more satisfactory results.

Acknowledgments. This study was funded by the National Key Research and Development Program of China (No. 2021YFB2600600), the Young Backbone Teacher Support Plan of Beijing Information Science & Technology University (YBT 202426), 2022 BISTU Scientific Research Fund (2022XJJ21).

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